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Acid Fracturing and Hydraulic Fracturing Applied in a Single Well for a Deep Carbonate Reservoir Appraisal; The Completion, Perforation, Stimulation and Data Gathering Experiences for 7 Zones

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Abstract

Acid Fracturing and Hydraulic Fracturing were used to stimulate and appraise a very challenging deep, low-porosity, carbonate reservoir with a complex combination of matrix background fractures (BF), and some large fault corridors and fault associated fractures (FAF). The well was drilled deviated through the reservoir to increase probability of intercepting fault corridors for production. The execution of the stimulations proved challenging due to the high strike-slip stress regime. Data from the pre-stimulation diagnostic tests was meticulously used to successfully plan and optimize each stimulation.

Acid stimulations on previous 4 wells showed challenges in formation breakdown as well as potential for shear-dilation pre-existing fractures during the injection phase. While during the production phase a rapid collapse of fracture conductivity was observed. Therefore, proppant stimulations were selected to 'preserve' the conductivity increase from the fracture dilation. For the larger fault zones, high leak-off was expected and acid fracturing was applied with emphasis on etching of the fracture surfaces and maximizing the acid diversion. Design preparations included laboratory conductivity testing of acid recipes and proppant on outcrop material. Pre-stimulation diagnostic tests were designed for the acid and the proppant fracs, considering the different reservoir characteristics of each stimulation target.

After the well was drilled, an integrated evaluation of mud logs, wireline logs, MPD drilling data and structural geology modeling was performed and 7 stimulation targets were identified. Acid stimulations were chosen for zones where faults and "fault-associated fractures" were identified and high leak-off was expected. The designs incorporated several diversion techniques that were successfully implemented to improve stimulation effectiveness. Proppant stimulations were placed in zones where only "background fractures" predominated, and initially followed designs typically pumped in unconditionals. The proppant stimulations, however proved to be the most challenging, and the pumping designs and perforation strategy were changed substantially after DFIT analyses. One hybrid stimulation treatment was conducted to create

an etched fracture and then tail in with a proppant treatment at the end of the stimulation. This practice is rarely attempted in the oil and gas industry but here the feasibility is demonstrated.

Intermediate clean-up and well testing were done to understand the reservoir characteristics and quantify whether acid fracturing or proppant fracturing is an appropriate stimulation method for the different zones in the reservoir.

A small group of 7 seismic stations were deployed to track the area natural seismicity and monitor the stimulation events. The quality of diagnostic information exceeded expectations, and the results could be linked to the performance of the stimulations. It showcases the feasibility of utilizing only a small array of geophones to diagnose the stimulation effectiveness.

Learnings and experiences for designing and execution of proppant fracturing, as an alternative to matrix or fracture acidizing in these complex carbonate formations are presented. Learnings for, successful diversion practices for acid fracturing in faulted carbonate zones with high leak-off are also shared.

Introduction

To date, five wells have been drilled in the appraisal field. Four of the wells have reached the Eocene to Cretaceous fractured carbonate reservoir (one well has failed to reach the reservoir due to drilling problems related to the complex stress regime). Well tests of varying duration exist for all four reservoir penetrations. The gas-condensate reservoir fluid contains H₂S (0.3 - 6.7 mol%) and CO₂ (0.4 – 2.1 mol%) and produced at GORs of 2400 – 3700 scf/stb. The wells have penetrated all structural domains from forelimb, crest to back-limb and to test the entire reservoir section (Fig. 1). Short-term well tests have been conducted in Well-1 and Well-2, and a short-term plus an extended well test were performed on Well-4. Well-5 penetrated the forelimb of the structure and is the deepest out of the five exploration/appraisal wells drilled (5623m TVDss), the only one to penetrate also the Lower Cretaceous.

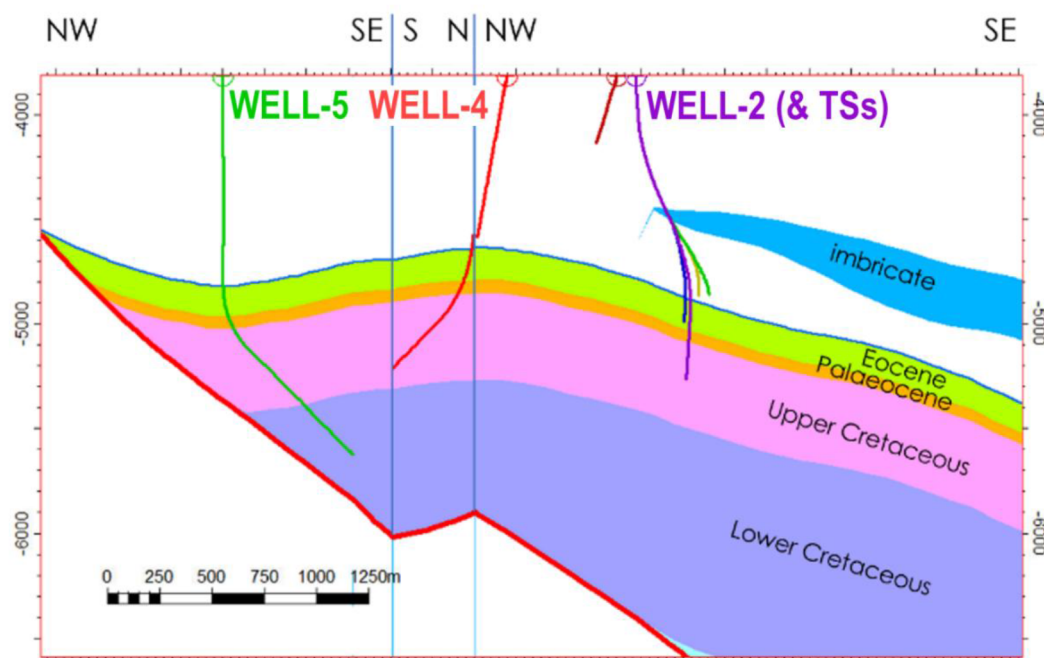


Figure 1—Well-5, Well-4 and Well-2 trajectories in cross section view, with Well-5 drilled at 45 deg deviation in the direction of the regional S_{hmin} and expected hydraulic fracture propagation perpendicular to the well.

Field Summary

The onshore field is a (thrust-)fault bounded three-way dip closure, and ranges over a length of ca. 26 km in north-south direction and ca. 6 km in west-east direction. The reservoir lies at a depth of 4-6 km

TVDss. The fractured carbonate reservoir consists predominantly of pelagic mudstones deposited during the Cretaceous to Eocene in a deep marine basin with occasional intercalations of calciturbidites of pack-to grainstone texture (<10%). Mineralogical logs and cuttings descriptions show that the rock is primarily limestone with occasional chert horizons. The matrix porosity of most samples ranges between 2-3%, with matrix permeabilities in the micro-Darcy to nano-Darcy range. The reservoir quality and productivity are mainly driven by the presence, abundance, conductivity, and connected volume of any naturally occurring fractures. Locally, solid bitumen in the reservoir can cause a reduction in reservoir quality but the quantity of pore-occluding bitumen in the reservoir is considered minor (<1%) with only 1 to 2 thin intervals of solid bitumen encountered in the wells.

On top of the fractured carbonate reservoir lies a marine shale (Oligocene Flysch) with some sand- and siltstone beds and few carbonate stringers. It forms the seal to the hydrocarbon accumulation.

Geomechanics

At present day, the region is in a strike-slip to compressional stress regime, (Heidbach et al, 2016), and the maximum horizontal stress ($S_{H,max}$) is oriented in a SW-NE direction. A few LOT data points are available in the field to constrain the minimum horizontal stress. Drilling experiences and the observed wellbore break-outs in the flysch overburden indicate a high horizontal stress anisotropy ratio in the range 1.2 to 1.6. The Flysch has a high clay content and contorted beds, creating borehole stability challenges. This has occasionally resulted in stuck pipe and/or lost tools requiring side-tracking (as in Well-3). To prevent swelling of reactive clays, the Flysch formation is drilled with oil-based mud. To successfully drill through the overburden, the strategy is to drill vertical and with high mud weights to counteract compressive shear failure and to stabilize low angle bedding planes.

The strike-slip stress faulting regime ($S_{H,max} > S_{v,vert} > S_{h,min}$) and the maximum horizontal stress direction is confirmed by break-outs observed in the reservoir section and well test observations. In the complex strike-slip faulting stress regime, many of the encountered fractures on the image logs are oriented such, that fracture reactivation pressure (FRP) required for shear-slippage occurs is lower than the minimum horizontal stress ($S_{h,min}$) (see Table 1). Evidence of shear-dilation was observed during drilling and stimulations on previous wells. Well-4 experienced losses into fractures through shear reactivation and eventually had total losses when a major fracture corridor was encountered within the reservoir section. Similarly, during the stimulation operations on Well-4, the Injectivity Index (II) showed an increasing trend while the Productivity Index (PI) showed a decreasing trend. The well test showed a depletion dependent productivity loss, which was explained by pressure dependent closure or aperture reduction of the created and preexisting natural fractures.

The absolute value of the present-day stresses is very high (see Table 1), with the vertical effective stress (VES) at top reservoir close to 49 MPa (~6,000 psi). Because the reservoir productivity is mainly driven by the fractures, preserving the conductivity of the shear fractures after stimulation is a concern as the effective stress is 51 MPa (~7,350 psi) at initial reservoir pressures and assuming 2000 psi drawdown pressure inside the fractures.

Table 1—Stresses at reservoir depth range

Depth (m TVDkb)	PP (MPa)	Sv (MPa)	VES (MPa)	Shmin (MPa)	Effective Stress (Mpa) with drawdown		SHmax (Mpa) for stress ratio		FRP (Mpa) for stress ratio	
					0 psi	2000 psi	1.2	1.6	1.2	1.6
4962	73	122	49	110	37	51	132	176	100	80
5765	78	141	64	126	48	62	151	201	114	92

Well-5: Drilling outcomes, Objectives and Stimulation Targets

Well-5 was drilled vertical through the overburden to top of the reservoir without encountering major stability and/or losses, proving the drillability of wells through the Flysch formation. The long reservoir section (1040 m AH; ~800 m TVD) was drilled deviated (45° inclination) to TD depth of 5623m TVDss. The well azimuth in the reservoir was in a south-eastern orientation with the aim of intersecting major fracture corridors in the reservoir. The reservoir section was drilled using managed pressure drilling technologies to reduce overbalance which allows better management of any potential mud losses. While drilling the reservoir section, no losses were encountered.

Wireline logging was run only over the top part of the open hole reservoir section. Due to significant number of events where the toolstring hung up and of high-tension pulls, the rest of the drilled section was logged on the drill string. Logging included VSP (cased hole), Resistivity, Sonic, GR, Caliper, BHI, and NMR (Fig. 2). There is no core or sidewall core data available from the reservoir to allow calibration of reservoir properties.

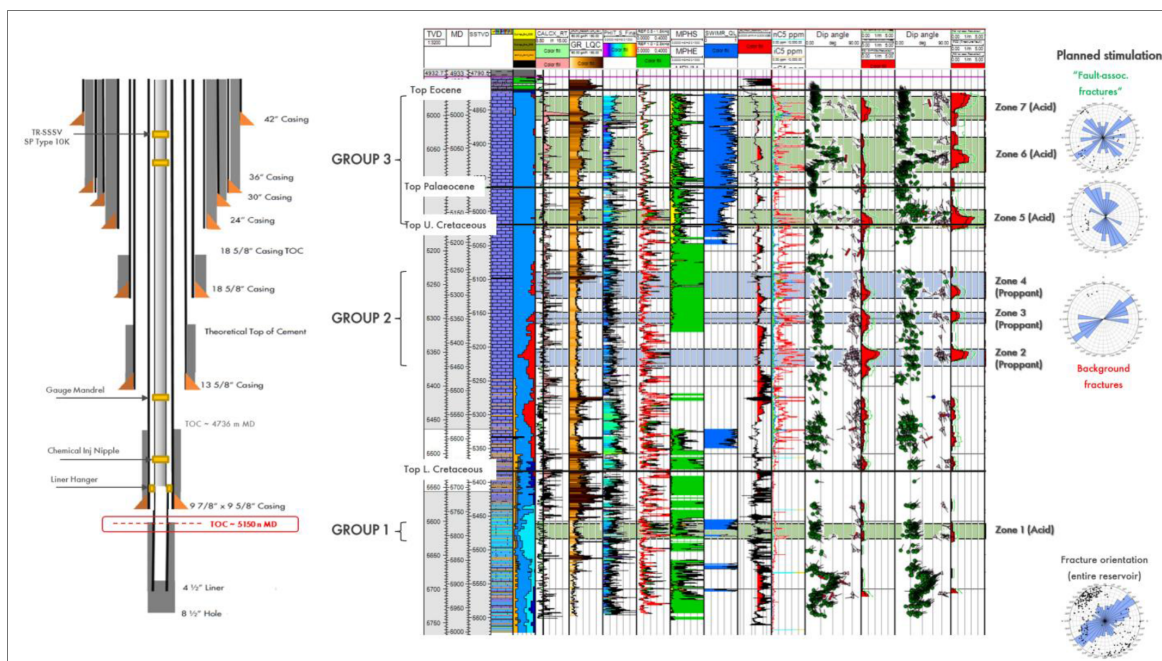


Figure 2—Well Completion with Logging Outcomes and Stimulation Targets

At the time of drilling Well-5, key subsurface uncertainties existed regarding the recovery per well, reservoir connectivity, background fracture conductivity, and matrix contribution to flow. The pressure build-up (PBU) interpretation after the extended well test on Well-4 suggested the permeability was deteriorating rapidly with pressure depletion. Producing from a fractured reservoir after acid stimulation, this was attributed to pressure dependent fracture closure i.e. fractures close with increasing drawdown. A major uncertainty remained regarding the efficiency of bullhead stimulation in the open-hole completion of Well-4 because the length of the open section was well over 600 m. Therefore, Well-5 was designed for a cemented liner completion (Fig. 2) which allows for plug & perf to test effectiveness of different types of stimulation, to attempt prevention of fracture closure and test effectiveness of various stimulation techniques using acid and proppant for different types of reservoir zones.

In total, 7 subsurface targets were identified for perforation, stimulation and well testing. Fig. 2 shows the interpreted identified faults and fractures intervals along the wellbore. Zones stimulated using the same technique would be grouped for a short individual clean-up flow test. Afterwards, all groups would be commingled in a final well test.

The cement-bond log (CBL) in the 4.5" liner showed that the top of cement (TOC) was much deeper than expected. Also casing wear was detected in 2 locations in 9 7/8 × 9 5/8" casing. Both issues impact the stimulation design & strategy. It lowered the maximum allowable stimulation pressures and required changes for the perforation and stimulation strategy for the upper 3 targets.

Selecting and Preparing Stimulation Designs for the Acid and the Proppant Zones

Rocks of the same age and depositional environment as the reservoir are exposed at surface in the region, and these outcrops allow for detailed analyses of the matrix and fractures. As can be seen in the plumbing diagram below (Fig. 3), the subsurface structure is quite complex and has an interplay of matrix, faults and fractures. Considering the subsurface uncertainties and based on the stimulation and test results from previous wells, it was decided that the project should have a "loss-zone-characteristics driven stimulation" concept i.e. combination of proppant and acid stimulations to cater to a wide range of subsurface scenarios. The main risks identified in the case of stimulation of a major loss zone i.e. a fracture corridor/faults with fault associated fractures, is that the leak-off could lead to screen-out for a proppant stimulation. Therefore, it was proposed to try proppant stimulations only in minor/medium loss zones characterized with total gas peaks, turbidite intervals and intervals with dense background fracture network.

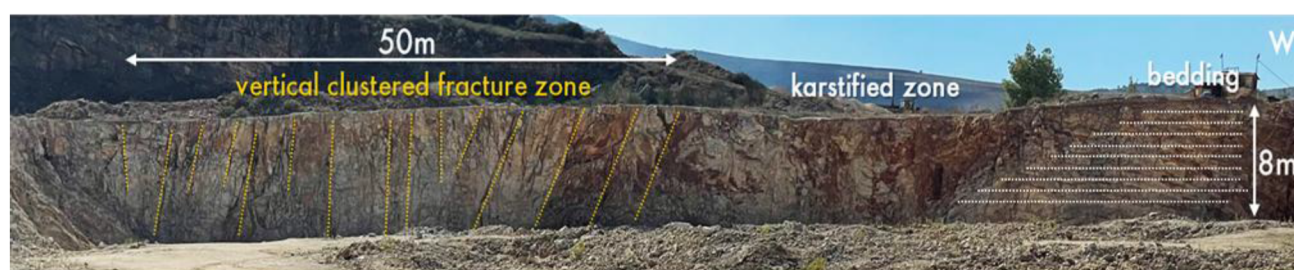


Figure 3—Outcrop Complex Subsurface Structure with Faults & Fractures

Acid Stimulation

The 'vertical clustered fracture zones' as shown in Fig. 3, were expected to have significant fracture permeability and high leak-off is to be expected. Therefore, acid stimulations were selected for zones 1, 5, 6 and 7, where the logs indicated high intensity of fault associated fractures. The objective of the acid stimulation is to reconnect the wellbore to the fracture system, bypassing the damage zone created by potential mud losses, completion fluids and cement. At the same time, the acid blend will be designed to create optimum etched conductivity using a self-diverting visco-elastic surfactant fluid system. Further diversion and leak-off is addressed in the pump schedule by adding both "small" and "large" degradable particulate material to act as diverting agents.

The success of the acid treatment depends on the created conductivity being retained under the high closure stress observed in the field. Retaining the conductivity is mainly a competition between two phenomena: the etching of the rock surface to create apertures and the collapse of these features under the high stress induced after the production drawdown is applied.

Different acid blends were tested on outcrop material to create acid etching patterns that promote the most sustainable conductivity. The acid blends and test conditions are shown in Table 2. It was not possible to run multiple conductivity testing due to difficulties in preparing competent samples and as a result there remains ambiguity here on the impact of experimental artefacts and in particular, loss of core competency during cutting, conductivity cell loading and core confinement. Fig. 4 shows the results of samples tested under stress loading in an API Fracture Conductivity Cell. Rock samples are subjected to increasing closure

stress over extended time. The tests capture three potential conductivity behaviors, which can be expected after the acid stimulation.

- Initial consideration of the results from Test 1 and Test 3, shows that the HCl will create wormholes and etched features that are highly conductive, and the decline seems almost flat i.e. the rock has more resistance to collapse. It should be noted that the 20% HCl in Test 1 was performed at much lower rates (10 ml/min) and acid exposure time was 3x higher than for the 28% acid. This likely led to an artificial result with a large wormhole/channel feature that extended from inlet to outlet of the core section and due to the strength and hardness of the core material, likely sustained to the highest stress tested here.
- Test 3 showed a wormhole/etching similar, but less extensive, than in test 1. Although Test 3 also had a very high initial conductivity, it showed collapse will occur at increasing stress state.
- Test 2 showed that the surfactant based self-diverting acid (SBSDA) (with 20% HCl concentration) will create the etched surface patterns along the targeted fracture planes. The SBSDA fluid created a more even dissolution face, and the fracture conductivity fell with closure stress more rapidly. The conductivity change with increased loading, shows the expected exponential decline. Note: Although the etched conductivity declines fast with increased stress the absolute final conductivity value (not shown) is still high.

Table 2—Acid Conductivity Test Specifications

Test #	Acid Blend	Injection Rate (ml/min)	Injection Time (min)
1	20% HCl	10	45
2	Surfactant Based Diverting Acid	50	15
3	28% HCl	50	15

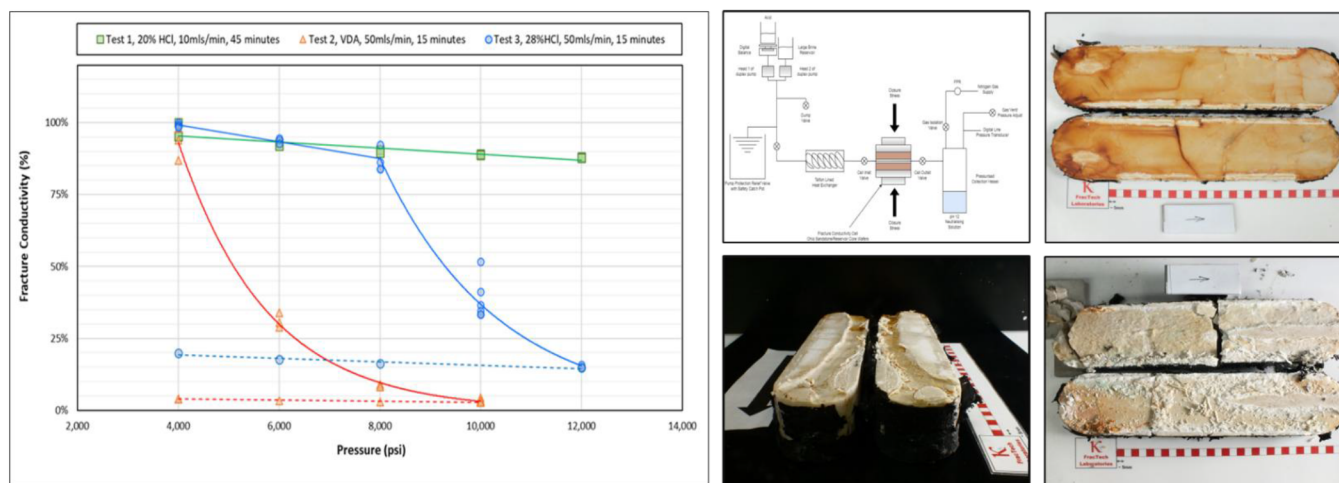


Figure 4—Stressed Conductivity Test Post Acid Injection: (a) Test results; (b) A schematic of the experimental set-up of the 10 square inch API conductivity cell; (c, d, f) samples after testing

At the end of the test the stress loading on the samples was released in steps, and all the test showed the conductivity loss due to loading is irreversible suggesting asperity crushing and other permanent deformation (see dotted lines in Fig. 4).

One of the challenges in performing conductivity tests for fracture acidizing is that the actual injection rates near the wellbore are quite high (40-100 fpm) and exceed the limits of the laboratory injection equipment by 1-2 orders of magnitude. All the test acids were pumped at significantly lower rates than

actual conditions which increases the exposure time of the acid to the core surfaces. Therefore, experimental results are likely to over predict the etched channel width, size of wormholes, and fracture conductivity especially at high temperatures. The test results were regarded as qualitative indicators rather than actual values that could be used for predicting production performance.

One thing is clear here and reproducible, that increasing core stress post acidizing induces a significant loss of conductivity (Abass et al. 2006). These observations were strongly considered when choosing either acidizing or propped fracturing treatments. Furthermore, the acid blends were thoroughly tested and the chemistry adjusted for compatibility with the hydrocarbon, the completion fluids and the frac fluid.

Proppant Stimulation

In typical hydraulic fracturing (HF) operations planar tensile fractures are created. However, in fractured reservoirs with high tectonic stresses, the hydraulic stimulation mechanism that is anticipated for this reservoir is that through interaction with the fracture network a stimulated rock volume (SRV) or stimulation domain will be created (Jamison, 2017)(Palmer, 2012). Th the applied over-pressure in the reservoir formation induces slipping of existing critically stress fractures or failure at weak bedding planes (McClure et. al.,2014). Likely a combination of both Mode 1 tensile failure and HS will occur (see Fig. 5). It should be noted that most existing hydraulic fracture modeling tools are insufficiently tailored and calibrated to design such fracs. Therefore, it is important to rely on analogues and experience to adjust the models and design the fracture treatments.

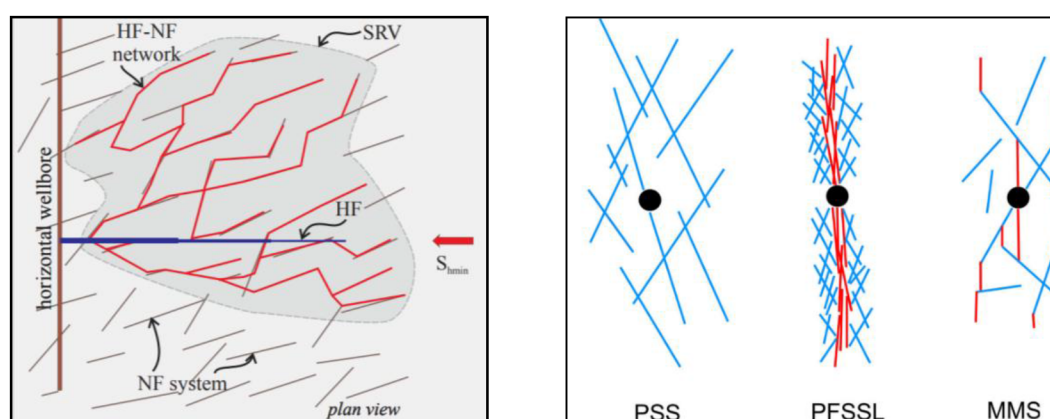


Figure 5—Conceptual models for the mechanism of stimulation (Left) how a hydraulic fracture (HF) might interact with the natural fracture (NF) system to create a dispersed HF-NF network or SRV (Jamison & Azad, 2017), and (Right) New fractures are represented with red lines, and preexisting fractures are represented with blue lines. The mechanisms are; pure shear stimulation (PSS), primary fracturing with shear stimulation leak-off (PFSSL), and mixed-mechanism stimulation (MMS) (McClure et. al.,2014).

Some design elements to use are a ‘thin’ & ‘clean’ fluid such as a high viscosity friction reducer (HVFR), and a fine-grained proppant that can penetrate and keep the existing fractures open. Analogue experience has shown that using conventional crosslinked fracturing fluids can be detrimental to well productivity. HVFR frac fluids have been successfully used in unconventional (North America) and deep tight reservoirs (Oman) and shown to create "cleaner and longer fracs" in these low permeability environments. The selected design was a ~40 MT stimulation using 100 mesh and a small tail-in with 40/70 frac sand pumped at 5-6 m³/min. Fracture simulations (not presented in this paper) indicated created fracture heights would be in the range of 35 to 40 meters, resulting in a proppant intensity of 1 to 1.5 MT/TVDm.

The effective stress on the proppant pack is well over 7,000 psi at initial in-situ stress conditions and assuming 2000 psi drawdown pressure inside the fracture during production. There is a concern for crushing and losing the conductivity of the proppant pack during production. Therefore, conductivity tests were performed to qualify the proppant using cores of Ohio sandstone and outcrop material. The results were also

used to set limitations for the bean-up strategy during the clean-up flow. Following the recommendations from Pearson et al. (2023), an additional small (10 % of total proppant volume) tail-in of ceramic proppant was included in the fracture design to further protect the (long-term) conductivity of the fracture, since the stress on proppant is highest at the near wellbore.

Perforation Strategies

Logs from Well-4 showed that fault zones, which are the possible major loss zone intervals, have heights ranging from 20 to 40 meters at the wellbore. To reduce the number of gun runs, the maximum perforated net height was limited to ~20 meters for the acid stimulations. In case the zone was larger than 20 meters, four perforated intervals of 5 meters would be distributed and spaced based on actual the log data.

The selected HVFR impacts the perforation strategy for the zones which would be hydraulically fractured. To keep the fractures wide enough and the fluid velocity high enough during injection to prevent a premature screen-out, the industry best practice was to use short perforated intervals of 1.5-2 meters (creating a low limited entry pressure of ~500 psi).

Natural Seismicity

Due to the active thrust tectonics, this region is seismically very active with several significant earthquakes reported in the last hundred years. Induced seismic activity in connection with injection or production of large volumes has been documented in many areas in the oil and gas industry. A seismic event was recorded during a bullhead acid stimulation on Well-4. The 2.1 magnitude event was estimated to have occurred at ca. ~2.4 km distance from the well.

Because of the sparse national network of seismic stations, it was decided to install a temporary seismic monitoring system around the area of interest. Best practices regarding monitoring seismicity monitoring were implemented for this well although they were not a local regulatory requirement. Five months in advance of the stimulation operations, 7 seismic stations were installed within a radius of 10km around the wellsite. This helped establish a baseline for the natural seismicity in order to compare to the seismic activity during the period of the Well-5 stimulations. The sensitivity of detecting a seismic event within the radius of 10km was magnitude 0.5 or larger. The events recorded within this (arbitrary) perimeter were termed "internal events". Most "natural" micro-seismic events were below a magnitude that could be felt at surface, with only one event above a magnitude 2.0 (magnitude 2.1 at a depth of ca. 2.1 km). Over the five months, a total of 39 internal events were recorded (ca. 2 internal seismic events per week) and confirmed the active natural seismicity in the area.

Automatic email alerts & text messages were sent real-time to relevant staff during the intervention activities e.g. perforating, stimulation, CT milling, clean-out, for any micro-seismic event of a magnitude 2.0 or larger (i.e., threshold for events that can be felt at surface). These could be used to allow for any operational changes if required, in ongoing activities and/or potential modifications to the stimulation designs to be pumped.

Acid Stimulation and Clean-up of Group 1

Group 1 included only Zone 1 in the Lower Cretaceous which was stimulated using acid.

Zone 1: Perforation and Pre-Stimulation Diagnostic Test

A pre-stimulation diagnostic test was successfully executed. However, the aspired brine injection rates could not be achieved because the treatment pressures reached the maximum allowable well pressure limits. There was no sign of pressure breakdown or shear fracturing and instead, a pressure roll-over was observed. After shut-in, the pressure decline did not reach estimated $S_{h,min}$ even after 12 hours suggesting a very tight

formation. The main concerns after the test were the high treatment pressure and no signature of a fault or fracture network.

Zone 1: Acid Stimulation

After pre-stim diagnostic test, no major changes were made to the planned stimulation design. The full acid stimulation was pumped into the formation including the large & fine degradable particulate stages. The stimulation rate was however constrained by the maximum allowable treating pressure of the well and planned pump rates could not be achieved. Fig. 6 shows the treatment pressure and rates, together with the recorded events. Notably, no pressure decrease was observed after the acid reached perforations. Only a few micro-seismic events were recorded towards the end of the stimulation and during pressure fall-off. The high treatment pressure suggested no significant shear-fracturing or fault reactivations.

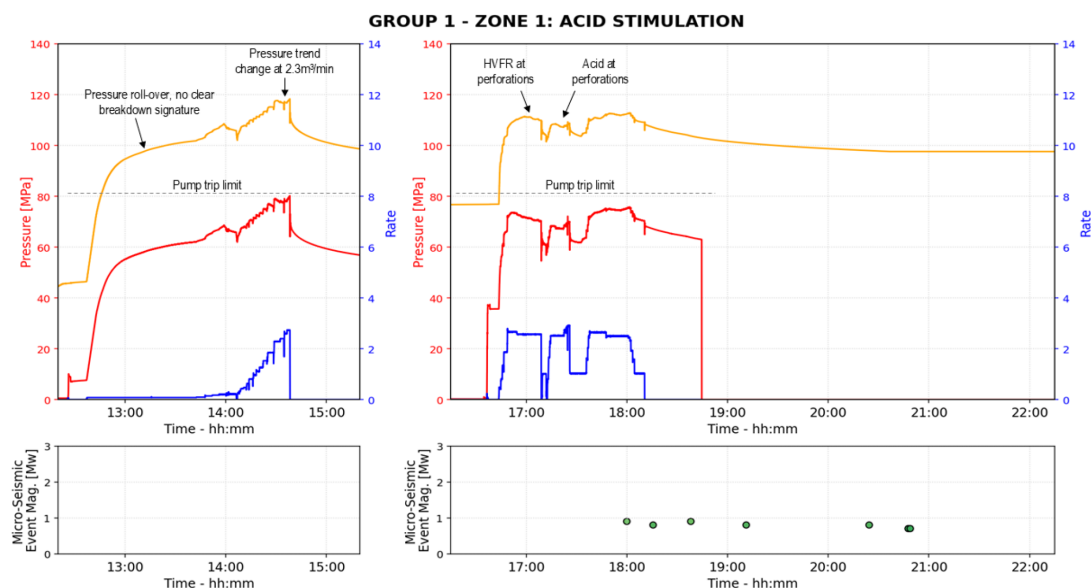


Figure 6—The top graph shows the pump data for the acid stimulation of Zone 1. The injection rate is shown in blue, and the surface treatment and bottomhole gauge pressure are shown in red and orange, respectively. The graph below shows the recorded micro-seismic events (dots). There is no correlation between the events and any pressure changes, suggesting there is no evidence of fracture or fault reactivations.

Group 1 Flowback

After the acid stimulation, the well was opened-up for clean-up. During this period, the liquid production was 100% spent acid (no formation water) and some gas that was produced at very low rates. Well-5 is the only well that penetrated and tested the Lower Cretaceous proving a deep hydrocarbon-water contact.

Proppant Stimulation and Clean-up of Group 2

Group 2 included Zones 2, 3 and 4 in the Upper Cretaceous. These zones were selected for stimulation of the back-ground fractures using proppant fracturing. The stimulation of Zone 3 was skipped because no breakdown of that formation could be achieved and the injectivity was too low to attempt a bullhead acid spearhead.

Zone 2: Perforation and DFIT

A 2 m interval with low stress (inferred from the Poisson's ratio) was identified and perforated. The diagnostic injection test (DFIT) showed no clear formation breakdown and instead a pressure roll-over (similar to Zone 1). was observed (Fig. 7). The injection pressure was high, and the test showed no signature of connectivity to a fault/larger fracture network. After the test, the concern was that the minimum rate for

pumping the proppant might not be achieved during stimulation. Therefore, the proppant treatment was redesigned for lower rate using lower proppant volume and concentrations.

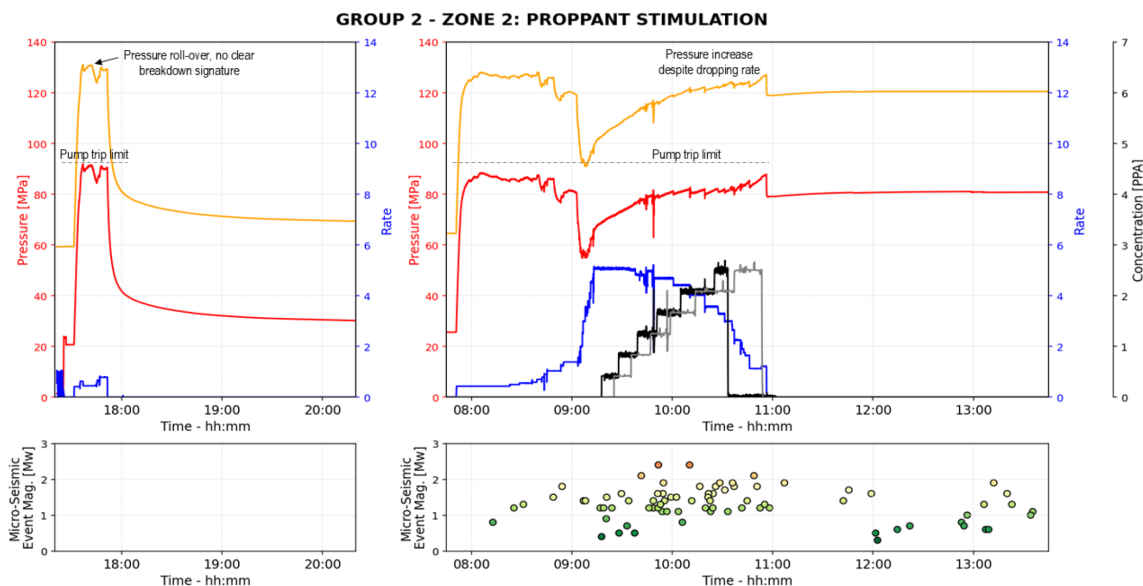


Figure 7—The top graph shows the pump data for the proppant stimulation of Zone 2, showing the injection rates, the surface treatment and bottomhole gauge pressure together with the recorded micro-seismic events (dots). There is a build-up of events that correlate with small pressure changes until the acid hit the perforations and the micro-seismicity increases coinciding with the large drop in treatment pressure indicating a connection with a fault and/or fracture network. Micro-seismic events continue also after the stimulation.

Zone 2: Proppant Stimulation

During the 1st hour of pumping, a few micro-seismic were measured correlating to the small drops in treatment pressure. However, much of the success for achieving higher injection rates depended on the effectiveness of acid spearhead to significantly decrease the treatment pressure. Fig. 7 shows the acid reached the perforations around 9:00 after which pressure shows a sharp drop. This allowed the injection rate to be increased and the start of the proppant placement. However, the pressure is seen to climb continuously even while pumping the clean pad prior to the proppant stages. The continual increase in the pressure led to a decision to terminate the treatment earlier than designed.

A "wave" of micro-seismic events is recorded during and after the stimulation. Although only 55% of planned proppant volume was injected, the 1st ever proppant stimulation in this field was considered successful. Higher than expected rates were achieved, but the learning was that the stimulation is injection volume-constrained, and unlike classical planar frac behavior the BHP rises continuously throughout the job even while lowering the injection rate. Similar behavior has also been observed in other very low permeability reservoirs.

Zone3: Perforation Strategy

Similar to Zone-2, 2 m of an interval with a low in-situ total stress was perforated. Following best practices from other fields, the perforation interval avoided any fractures identified on borehole image log (BHI) in order to reduce the risk of screen-out. Yet, no formation breakdown or injection could be achieved in zone 3.

Prior to a second stimulation attempt, the perforation strategy was revised, and it was decided to perforate an additional 5 m across an interval with natural fractures. The estimated stresses were higher than in the previous perforated interval but included some thin intervals which showed a decrease in the Poisson's ratio. Again, no breakdown was achieved, despite attempting multiple breakdown cycles and forced pressure fall-offs. The injectivity was insufficient for acid or proppant stimulation. It was decided not to use the contingency option to spot acid with coiled tubing (CT) and to skip the stimulation on zone 3.

Zone 4: Perforation and Injectivity Test with Acid Pill

Following the outcome on Zones 2 and 3, the perforation strategy, pre-stimulation diagnostic test, and stimulation design for Zone 4 were revised. The priority shifted towards establishing connectivity to the reservoir and maximizing injectivity i.e. regret the data collection objectives and accepting the increased risk of screen-out by perforating a larger interval and include fractures. A 10 meter interval was perforated and for the 1st time also under-balanced perforating was applied. The pressure response following the underbalance perforating indicated a good connection with the reservoir.

The ‘injection test’ showed a pressure roll-over instead of a clear breakdown (Fig. 8), similar to Zone 1 and Zone 2. As soon as the injection rate was reached, an acid pill was launched. When the acid reached the perforation but showed no clear signs of any response. No distinct fracture closure was observed during fall-off. As in the previous stages, the main concern after the test was the high treatment pressure and a low injectivity rate for a proppant stimulation.

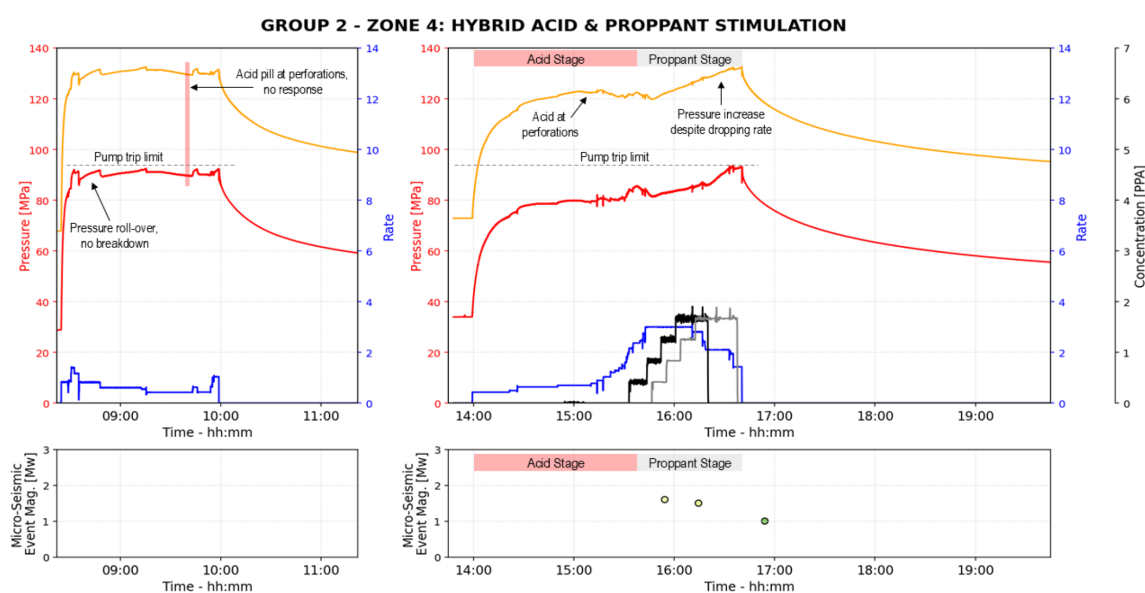


Figure 8—The top graph shows the pump data for the proppant stimulation of Zone 4, showing the injection rates of the acid and HVFR hybrid stimulation. The lack observed micro-seismicity (three dots in the bottom graph), and the continuous increase in treatment pressure suggests the stimulation is not connected to any fracture network. The acid stimulation was crucial to achieve the minimum required rate to start placing the proppant.

Zone 4: Hybrid Stimulation

The stimulation design for Zone-4 was changed to a hybrid stimulation of a large acid stage ahead of the proppant stimulation. The main job started at lower injection pressure than expected suggesting that perhaps the acid pumped during the earlier diagnostic test may have had a delayed benefit. Surprisingly, the larger acid stage in the main treatment did not create further pressure reduction (Fig. 8). Nonetheless, there was sufficient room to increase the rate prior to the start of the proppant placement. However, as in the previous zones, the treating pressure continued to increase with injected volume. Although not all the planned proppant could be placed, the feasibility of the hybrid stimulation treatment was demonstrated in a complex stress regime.

Compared to zone 2 only a limited number of micro-seismic events were registered.

Group 2 Flowback

After stimulation of Zone-4, all the plugs between Zones 2 to 4 were milled and Group 2 was flowed for the clean-up.

The well produced mainly water (spent acid and stimulation fluid, no formation water) and some gas. The flow was slugging and started to die after ~20 hours. Nitrogen lift via coiled tubing was attempted to offload the well, but this did not result in an increase of inflow from the formation and was stopped after 10 hours. In total the well recovered ~20% of injected stimulation fluid and spent acid.

Stimulation observations point towards low permeability expected from Zone 4 due to the lack of micro-seismic events and high treating pressures. Clean-up flow most likely was only from Zone 2 and confirmed the low permeability with limited network of background fractures. At the end of the flowback period, a stable flow was achieved, and the well productivity was insufficient to produce back the large volumes of stimulation fluid injected during the proppant stimulation.

Acid Stimulation and Clean-up of Group 3

Group 3 included targets in the Paleocene and Eocene and was stimulated with acid.

Zone 5, Zone 6-7: Perforation

The BHI log showed evidence of high natural frac density in Zones 6 and 7, and those were most promising zones for production. However, the CBL showed the top of cement was below Zone 6 (Fig. 2) i.e. there was no isolation between Zone 6 and 7. The risk was that the acid would flow through the open annulus and only preferentially stimulate only one of the zones. The best chance of stimulating both zones (6 and 7) was to try dynamic diversion by reducing the number of perforations to create a high velocity jetting action right across Zones 6 and 7. The high velocity jet could help to initiate an acid wormhole at the desired location.

The perforation strategy for Zone 5 also was revised and included a stand-off height from the cement top to avoid compromising the annular cement isolation between Zone 5 and the uncemented portion of the liner (i.e. Zones 6 and 7).

Zone 5 Pre-Stimulation Diagnostic Test

Zone 5 in the Palaeocene was perforated under-balanced but showed no pressure response. The pre-stim diagnostics were performed without problems (Fig. 9), but again no breakdown was observed, and the BHP was continuously rising throughout the test. No pressure stabilization was observed during the test.

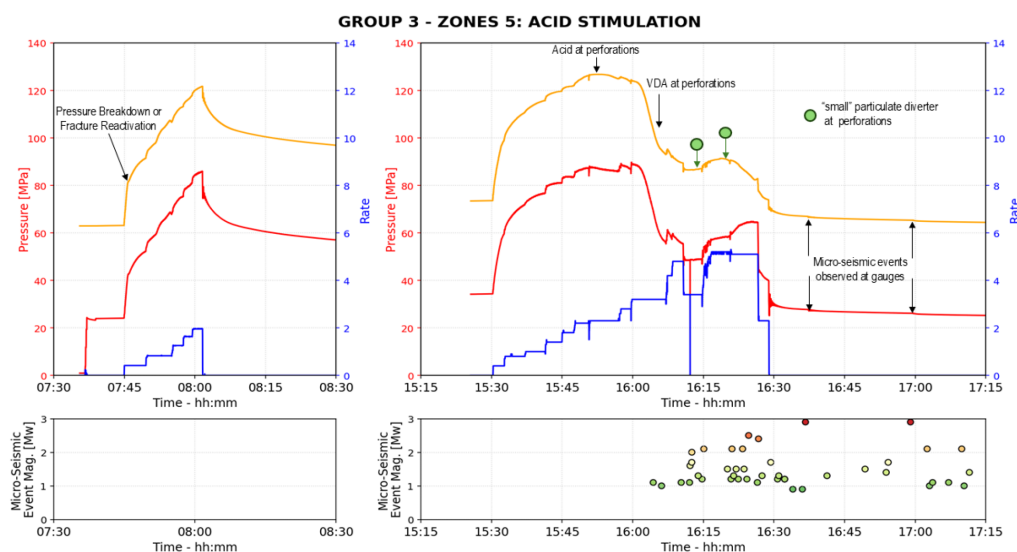


Figure 9—The top graph shows the pump data for the acid stimulation of Zone 5, showing the injection rates (blue), the surface treatment pressure (red) and bottomhole gauge pressure (orange) together with the recorded micro-seismic events shown in the bottom graph. The onset of the micro-seismic events coincides with the large drop in treatment pressure indicating a connection with a fault and/or fracture network. Micro-seismic events continue for more than two hours after the pumps shutdown. 24 induced micro-seismic events recorded, and 40 micro-seismic events recorded after the stimulation.

Zone 5: Acid stimulation

Following the pre-stim diagnostic test, the acid stimulation was successfully pumped into the formation. Zone 5 was the first interval where injectivity was not a concern, and stimulation was pumped at the max available pump rate. In total ~120 m3 of acid was pumped with clear signs of the acid stimulating a fracture network (Fig. 9). The large pressure drop, and onset of the micro-seismic events clearly show the connecting to a large fracture network. Not enough pump rate was available to keep the injection pressures above the (estimated) fracture opening pressure. There were 24 induced micro-seismic events recorded during pumping, and 40 events were recorded after the stimulation. The 2 large events ~16:35 and ~17:00 were also observed on the downhole gauges.

Zones 6+7: Pre-Stimulation Diagnostic Test

The Diagnostic Fracture Injection Test (DFIT) on Zone 6 & 7 showed low treatment pressure at low rates and almost no change in treatment pressure when increasing rate. This suggested that Zone 6 and/or 7 were connected to a large open network.

Zone 6+7: Acid Stimulation

During the stimulation, 170 m3 of 28% HCl Acid and 4 sweeps VDA-20% HCl with "small" particulate diverter stages were successfully pumped with clear signs that diversion was working. The surface rate had to be dropped to inject the diverter stages. After each diverter stage reaches the perforations the treatment pressure increases, showing how the "small" degradable materials help divert the acid into non-stimulated fractures. Stimulation of Zones 6&7 was the second stimulation stage after Zone 5 with no concerns regarding injectivity. Low injection pressures were observed at the maximum available acid injection rates (Fig. 10). Low treatment pressure and early occurrence of micro-seismic events suggest this stage is connected to a fracture/fault zone from the start, and possibly the same feature from Zone 5. Again, significant number of events were recorded after the stimulation and one large event ~15:00 was also observed on the downhole gauges.

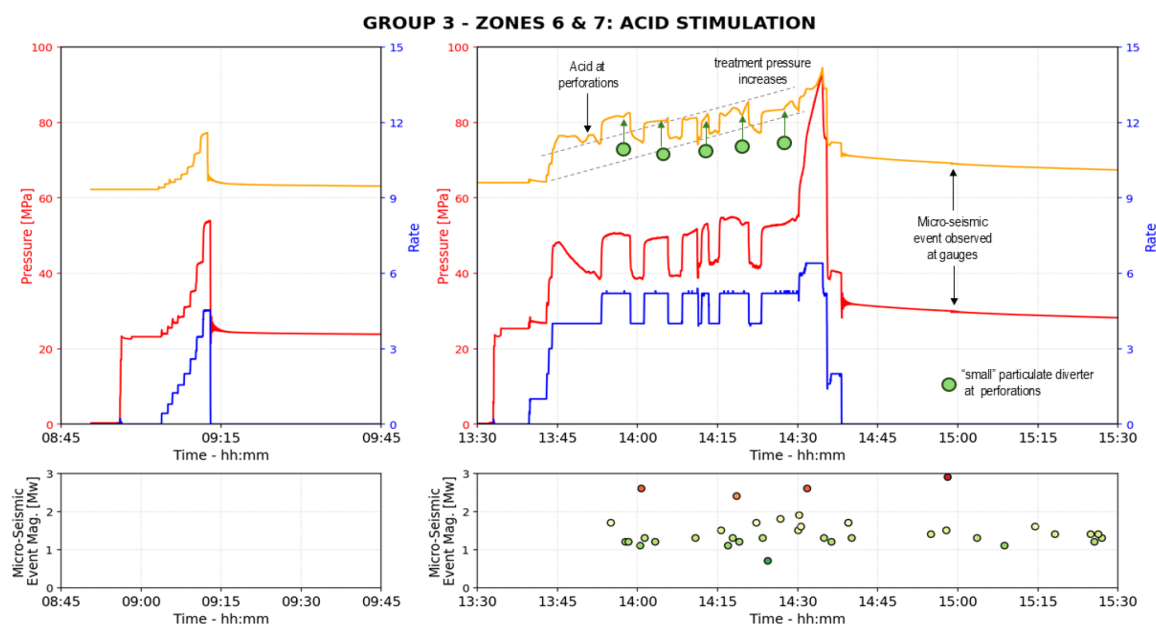


Figure 10—The top graph shows the pump data for the acid stimulation of Zone 6 & 7, showing the injection rates (blue), the surface treatment pressure (red) and bottomhole gauge pressure (orange) together with the recorded micro-seismic events shown in the bottom graph. Clear indications that diversion was working. Low injection pressures at the maximum available rates of 4 to 5.2 m3/min.

Group 3 Flowback and Commingled Test

The clean-up flow produced fluid was mainly oil and gas. Water was produced in slugs sporadically reaching up to 20% water-cut, but most of the water was produced during the first slugs, and after that the average water-cut was less than 3% (only spent acid, no formation water was observed). After 3 days of clean-up flow, all the plugs were milled, and the well was put into commingled production from all groups. No PLT was run to confirm flow contributions of Group 2 and 1.

Because Group-1 was the most promising zones for production, there was no expectation that Group 1 and 2 would start to contribute to the flow in the commingled flow. However, the production from these zones could help offload the fluid column and allow clean-up of the remaining stimulation fluids. Ultimately 56% of all the fluids pumped was recovered by the end of the test.

Induced Micro-Seismicity During Stimulations

Over the period of the Well-5 stimulation and well testing (ca. 6 weeks), a total of 354 internal micro-seismic events were recorded. The increase in the frequency of the events can be strongly correlated to the individual phases of the well stimulation. The micro-seismic events occur over an interpreted depth interval of ca. 4800 to 6300m TVDss, which puts them consistently within the reservoir interval suggesting that the injection of stimulation fluids induced micro-seismicity in the reservoir. The observed magnitudes are indicative for the reactivation of natural structures. Almost all the events were very low magnitude that would be undetectable at surface.

Although, uncertainties exist in the simplified 1D velocity model and in the depth conversion used to locate the events, it can be assumed that the true location of the micro-seismic events are all adjacent to the Well-5 wellbore.

The occurrence of the micro-seismic events during stimulation, the final geometry of the cloud of micro-seismic events, and the trend of the treatment pressure throughout the stimulation jobs, has helped explain the effectiveness of the stimulations in the different zones and the geological features e.g., background fractures versus fault associated fractures. Three distinct ‘geometries’ were observed, a) quiet zones with few events, b) clustering of events around the wellbore and c) an elongated stretch of events.

The Quiet Zones (Zones 1 and 4)

The picture on the left (A) in Fig. 11 shows the events recorded during stimulation of Zone 1 (Note: Zone 4 looks similar). Both stimulations on Zones 1 and 4 had no pressure drop when acid hit the formation and had high-treating pressures that increased with total injected fluid volume. Rates had to be dropped continuously during the treatment as the pressure was rising and close to the pressure limitations of the well. Although the treatment pressure and the placement of large amounts of proppant in Zone 4 confirm that fracturing took place, the lack of micro-seismic events suggest that no fracture network existed and/or was activated by the injected fluids. Treatments in these zones were more likely to be classical Mode 1 or planar fractures.

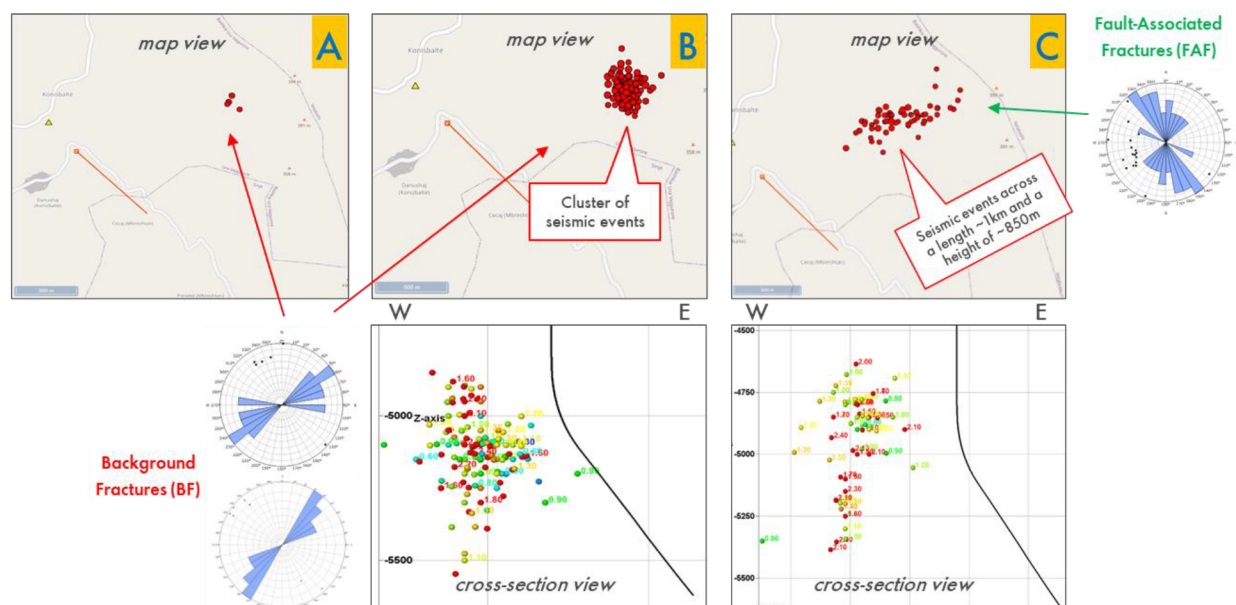


Figure 11—The micro-seismic events presented in map view and side view, on the left (A) shows the few events recorded during stimulation of Zone 1, in the middle (B) shows micro-seismic events clustered around the wellbore for zone-2, on the right (part C) shows the micro-seismic events form an elongated cluster over a length of ca. 1,000m, in an $\sim 30^\circ$ angle to the regional maximum horizontal stress direction SW-NE pointing towards a potential shear reactivation.

Events Cluster Around the Wellbore (Zone-2)

Large number of micro-seismic events were recorded during injection into Zone-2. Some events can be directly linked to changes in the treatment pressure. The picture in the middle (B) of Fig. 11 shows the events form a cluster in 3D around the well. This is consistent with the creation of a stimulated rock volume without signs of a planar frac development.

The Proppant fracture on Zone 2 showed micro-seismic events from the start of stimulation. The timing and clustering of events and rising treatment pressure suggests connection and stimulation of a network of background fractures (BF), likely through shear stimulation. Proppant could successfully be placed inside the background fractures. However, as evident from the near wellbore events and the increasing pressures, the formation stresses were increasing possibly due to poro-elastic stiffening of the surrounding rock.

A Fracture Interacting with Natural Fractures (Zone 5 and Zone 6+7)

The picture on the right (C) in Fig. 11 shows a clear example where the acid stimulation in zone 5 connected with a pre-existing large fracture network. This is observed both by the drop in treatment pressure and the onset of the micro-seismic events.

The geometry of the band of micro-seismic events in map view shows that the direction of the events is oriented $\sim 30^\circ$ to the regional maximum horizontal stress direction over a length of 1000m confirming the shear-reactivation and stimulation of a pre-existing fault/fracture corridor. Mapping of the micro-seismic events indicate that an ~ 1000 m long fracture/fault zone was stimulated. A similar geometry of micro-seismic events for the combined stimulation of Zone 6 and 7 was observed. The events from Zone 6+7 are almost overlapping with the location of the events from Zone 5. The 'cloud' of Zone 6 and 7 micro-seismic events did show differences in location and orientation compared to Zone 5. Recognizing that previously activated features remain 'quiet' since they are already dilated and thus are unlikely to cause another event,, it is highly likely that the stimulation reconnected with the same fracture system and fault. This also is in-line with the low treatment pressure and the observation that onset of events starts almost at the beginning of the early start of the stimulation.

Project Highlights, Lowlights and Learnings

Extremely challenging stimulations were performed in a fractured tight carbonate in complex strike-slip stress faulting regime and high tectonic stresses. All the acid treatments including diverter pills were pumped as per plan. The proppant stimulations were pumped without any screen-out's which is a significant achievement in this reservoir, recognizing it was the 1st time proppant stimulations were performed in this field.

The success of this project was due to the integrated teamwork of the Subsurface, Central Design, Completions and Wells teams. Early involvement and collaboration of both in-house SME's and the service provider helped maintain flexibility in the stimulation designs and develop contingencies for expected/unexpected challenges. During the stimulations, with live support from PTE/SMEs during the job, the service provider and Shell acted as one team.

The test on the outcrop material provided useful insights regarding rock strength, and the results showed that similar conductivity declines under stress for acid and 100 mesh frac sand. In general, softening of the asperities on the fracture surfaces is expected to cause a rapid conductivity decline. Therefore, it has been proposed to use proppant stimulation to keep the fractures open (Abbas et. al. 2006). However, frac conductivity testing is recommended if the rock strength is very competent to assess if acid fracturing is also viable under high stress conditions. For this hard rock reservoir, the conductivity decline of the acid etched fracture surfaces is similar to that of propped fractures.

In the stimulation designs it was assumed that a combination of both tensile failure (HF) and 'domain stimulation' or hydro-shearing' (HS) would occur. This was confirmed by the mapping of the micro-seismic events. And although the existing frac modeling tools insufficiently simulated or fully matched such fracture development, both in-house and service provider simulation models identified the potential for challenges related to high treatment pressure for the proppant stimulations. The HVFR fluid was critical to place proppant within pressure limitations.

Seismic monitoring delivered an improved understanding of subsurface response to the stimulations. It demonstrates that the risk of large seismic events can be managed for stimulations even in a highly active tectonic area. The micro-seismic events measured during the pumping of stimulation fluids gave insights in the stimulated rock volumes and in the fracture orientation (there were no micro-seismic events during diagnostic injections). The increase of the number of events correlated with the pressure drops in the treatment, and the lack of events with the challenges to maintain pumping below the pressure limitations. Micro-seismic events continue for many days after stimulation with decreasing frequency.

The concentrated cloud of events around the wellbore for Zone 2, showed evidence of stimulations of fractures in all directions around the wellbore, and is also suggested by the observed increasing pressure response that is correlating more to the total injected fluid volume rather than the proppant concentration. In this reservoir, in intervals with a network of only background fractures, there may be limitations on the total size (volume) of the fracture treatment that can be successfully placed.

To maximize the effectiveness of the acid stimulations required incorporating both a surfactant based diverting acid, conventional HCL, and degradable diverting agents in the stimulation design. The acid stimulations executions were designed for a maximum rate 5 m³/min, (using the same equipment setup as the proppant stimulations) and this limitation proved to be too low to keep-up with the leak-off. Higher pump rate for zones with fault associated fractures may be beneficial to increase the pressure above the shear fracturing pressure and open-up a larger network. The fine PLA diversion treatments deployed on zone 6 and 7 temporarily decreased the injectivity index, forcing treatment fluid into other areas of the extended natural fracture network and helping to control leak off.

On Zone 4 a hybrid stimulation treatment was conducted to create an etched fracture and then tail in with a proppant treatment at the end of the stimulation. This practice is rarely attempted in the oil and gas industry but here the feasibility is demonstrated. It is believed this would be very beneficial to develop deep

carbonate reservoirs with extremely high stress and closure pressures, negating etched features at least at the near wellbore from collapsing during post closure, clean up and with high drawdowns when the well is placed on production.

During the period in which the subsurface work, drilling and testing was conducted, the costs for drilling deep land wells increased by tens of percent and further appraisal activity was cancelled.

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References

- Pearson, M.C., Green, C.A., Milton-Taylor, D. and Trevino, V. (2023) "Extended-Time Conductivity Testing of Proppants at 0.5 lb/ft² Used for Slickwater Multi-Stage Horizontal Completions", SPE-212334-MS, SPE Hydraulic Fracturing Technology Conference and Exhibition, The Woodlands, Texas, January 2023. doi: <https://doi.org/10.2118/212334-MS>.
- Abass, H.H., H.H., Al-Mulhem, A.A., Alqam, M.S. and. Mirajuddin, K.R. (2006) "Acid Fracturing or Proppant Fracturing in Carbonate Formation? A Rock Mechanic's View", SPE-102590-MS, SPE Annual Technical Conference and Exhibition held in San Antonio, Texas, U.S.A., 24–27 September 2006. doi: <https://doi.org/10.2118/102590-MS>.
- McClure, M.W., Horne, R. N. (2014) "An investigation of stimulation mechanisms in Enhanced Geothermal Systems", *International Journal of Rock Mechanics and Mining Sciences*, Volume 72, pages 242-260, doi: <https://doi.org/10.1016/j.ijrmms.2014.07.011>.
- Suzhen, Li, Yun, Xu, Yujin, Wan, Mingguang, Che, Liao, Gao Yuebin, Zhanwei, Yang, Rui, Gao, Xinwei, Lu, Lang, Zhou, Yezhong, Wang, Weihua, Chen, and Zhou Hao. "Hybrid Multi-Stage Acid-Fracturing in Deep Carbonate Gas Wells with HTHP in Anyue Gas Field of Sichuan Basin, SW China." Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November 2017. doi: <https://doi.org/10.2118/188715-MS>.
- Ren, Jichuan, Yao, Ziqiang, Guo, Jianchun, Xu, Peixuan, Jie, He, Zhang, Tao, and Yan Deng. "Constructing Flow Capacity of Fracture Networks in Deep Carbonate Reservoir Through Hybrid Acid Stimulation." Paper presented at the SPE Conference at Oman Petroleum & Energy Show, Muscat, Oman, May 2025. doi: <https://doi.org/10.2118/224899-MS>.
- William Jamison, Ali Azad, The hydraulic fracture – natural fracture network configuration in shale reservoirs: Geological limiting factors, *Journal of Petroleum Science and Engineering*, Volume 159, 2017, Pages 205-229, ISSN 0920-4105, <https://doi.org/10.1016/j.petrol.2017.09.017>.
- Palmer, Ian D., Moschovidis, Zissis A., and Aaron Schaefer. "Microseismic Clouds: Modeling and Implications." Paper presented at the SPE Americas Unconventional Resources Conference, Pittsburgh, Pennsylvania USA, June 2012. doi: <https://doi.org/10.2118/154903-MS>.
- Heidbach, Oliver; Rajabi, Mojtaba; Reiter, Karsten; Ziegler, Moritz; WSM Team (2016): *World Stress Map Database Release 2016*. V. 1.1. GFZ Data Services. <https://doi.org/10.5880/WSM.2016.001>.
- Zhang, F., Damjanac, B. & Maxwell, S. Investigating Hydraulic Fracturing Complexity in Naturally Fractured Rock Masses Using Fully Coupled Multiscale Numerical Modeling. *Rock Mech Rock Eng* 52, 5137–5160 (2019). <https://doi.org/10.1007/s00603-019-01851-3>.
- Smith, Michael B, Deng, Jia Yao, Sajid, Bilal, Mokhtar, Ahmed Anwar, and Musa Taramov. "Simulation, Design and Application of Acid+Proppant Stimulation in a Carbonate Reservoir." Paper presented at the SPE International Hydraulic Fracturing Technology Conference and Exhibition, Muscat, Oman, October 2018. doi: <https://doi.org/10.2118/191409-18IHFT-MS>.